Homework 6: Regression

Submission Notes: Please write your answers to each question under that question (not altogether at the end). It will be easier for me to give full credit if your answer is commented well, including comments about where you are completing each sub-part of the question. For an example, see Homework 1.

---

1. In Lecture 13, we discussed the linear algebra formulation of linear regression. For example, we can set up perfectly simulated regression data as follows:

```
simulate.reg <- function(n=100,
                         betas=c(4.35, 1.73, 0.02, 2.78),
                         sigma=4.16,
                         set.seed=F){
  # Specify Coefficients
  beta <- matrix(data = betas, nrow=4)
  # Specify Error Distribution
  mu.epsilon <- 0
  sigma.epsilon<-sqrt(sigma)
  # Simulate Regression Data
  if(set.seed){set.seed(7272)}
  x1 <- runif(n=n, min=-5, max=5)
  x2 <- runif(n=n, min=-5, max=5)
  x <- cbind(1, x1, x2, x1*x2)
  epsilon <- rnorm(n=n, mean=0, sd=sigma.epsilon)
  y <- x %*% beta + epsilon
  # Return simulated data
  dat <- tibble(y=y, x1=x1, x2=x2)
  return(dat)
}
```

   In Lecture 15, we discussed regression-based inference. During office hours, I had some discussions about the mathematical statistics behind these results. Although we don't do the real-analysis style derivations in this course that we might in Mathematical Statistics, we have plenty of tools to investigate these ideas numerically. Complete the parts below to conduct this exploration.

   (a) Use the `simulate.reg()` function to simulate regression data using the default set of coefficients over a grid of σ_ϵ values from $\sigma_\epsilon = 0$ to $\sigma_\epsilon = 5$. Compute the SS_{res} and SS_{reg} from the resulting regression model, and plot the points using color to denote the σ_ϵ value. Make sure you set `set.seed` to `T` so you can compare the output at fixed $\mathbf{x}$ and $\mathbf{Y}$. **Why do you see the pattern that emerges?**
   Hint: I found it was easier to manually compute SS_{res} and SS_{reg} than it was to extract them from the result of an `aov` call.

```r
asim <- tibble(sigma = rep(NA, 1000),
           SSres = rep(NA, 1000),
           SSreg = rep(NA, 1000))

sigmas <- seq(from = 0, to = 5, length.out = 1000)

for(i in 1:1000){

  dat <- simulate.reg(sigma = sigmas[i], set.seed=T)

  asim$sigma[i] <- sigmas[i]

  anova <- aov(y ~ x1 + x2 + x1*x2, data = dat)

  asim$SSres[i] <- sum((dat$y - anova$residuals)^2)

  asim$SSreg[i] <- sum((anova$residuals - mean(dat$y))^2)

}

ggplot(data = asim, aes(x = SSres, y = SSreg, color = sigma)) +
  geom_point() +
  xlab(bquote(SS[res])) +
  ylab(bquote(SS[reg])) +
  labs(color = bquote(sigma[epsilon])) +
  theme_bw()
```

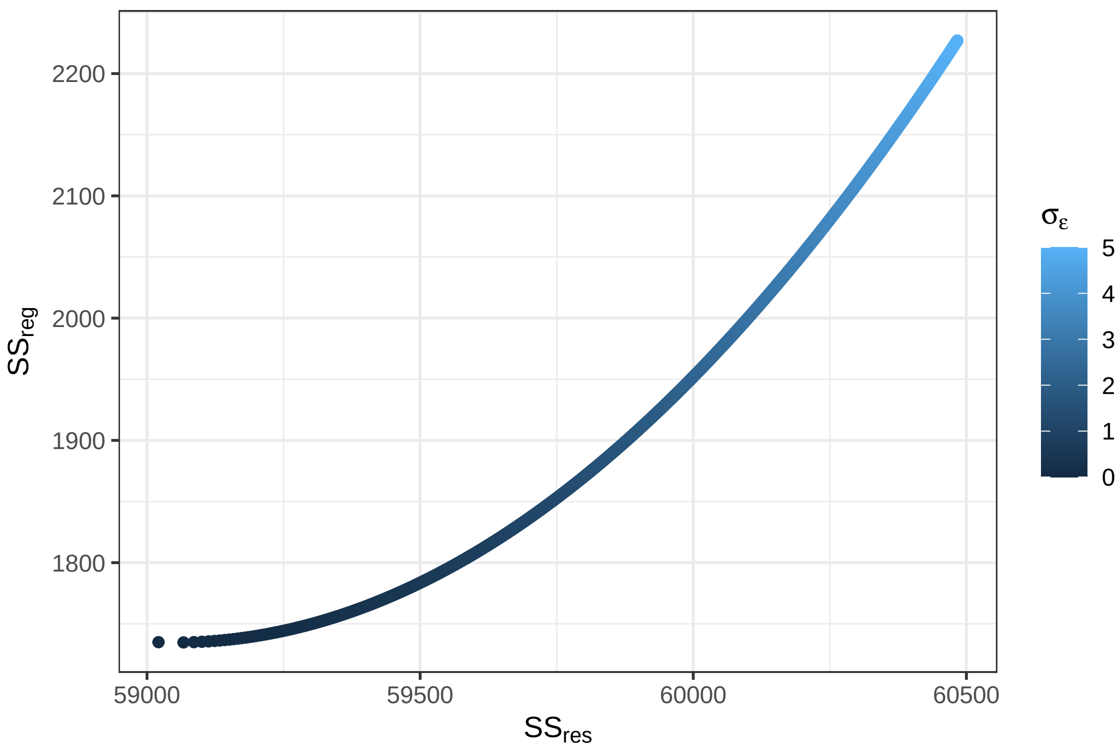

Figure 1: σ across levels of SS_{res} and SS_{reg} (fixed $\mathbf{x}$ and $\mathbf{Y}$)

We see that higher values of SS_{res} and SS_{reg} are associated with higher values of σ_ϵ. This makes sense as σ_ϵ affects the variation around how the data fits the model, and higher variation in σ_ϵ will result in higher levels of variation in the regression residuals and against $\bar{Y}$.

(b) Use the `simulate.reg()` function to simulate regression data using the default set of coefficients. Create a plot that shows the distribution of the βs. Make sure you set `set.seed` to `F` so you can compare the output for differing samples of $\mathbf{x}$ and $\mathbf{Y}$. **What do you notice about their distributions?**

```
bsim <- tibble(Beta0 = rep(NA, 1000),
           Beta1 = rep(NA, 1000),
           Beta2 = rep(NA, 1000),
           Beta3 = rep(NA, 1000))

for (i in 1:1000){

  dat <- simulate.reg(set.seed=F)

  mod <- data.frame(summary(lm(y ~ x1 + x2 + x1*x2, data = dat))$coefficients)

  bsim$Beta0[i] <- mod$Estimate[1]
  bsim$Beta1[i] <- mod$Estimate[2]
  bsim$Beta2[i] <- mod$Estimate[3]
  bsim$Beta3[i] <- mod$Estimate[4]


}

bsim <- bsim |>
  pivot_longer(cols = colnames(bsim),
           names_to = "Beta",
           values_to = "Value")


library(ggridges)
ggplot(data = bsim, aes(x = Value, y = fct_rev(as.factor(Beta)))) +
  geom_density_ridges(alpha = 0.5) +
  theme_bw() +
  xlab("Estimate") +
  geom_hline(yintercept = 0) +
  ylab('') +
  scale_y_discrete(
labels = c(
  "Beta0" = bquote(beta[0]),
  "Beta1" = bquote(beta[1]),
  "Beta2" = bquote(beta[2]),
  "Beta3" = bquote(beta[3])
)
  )
```

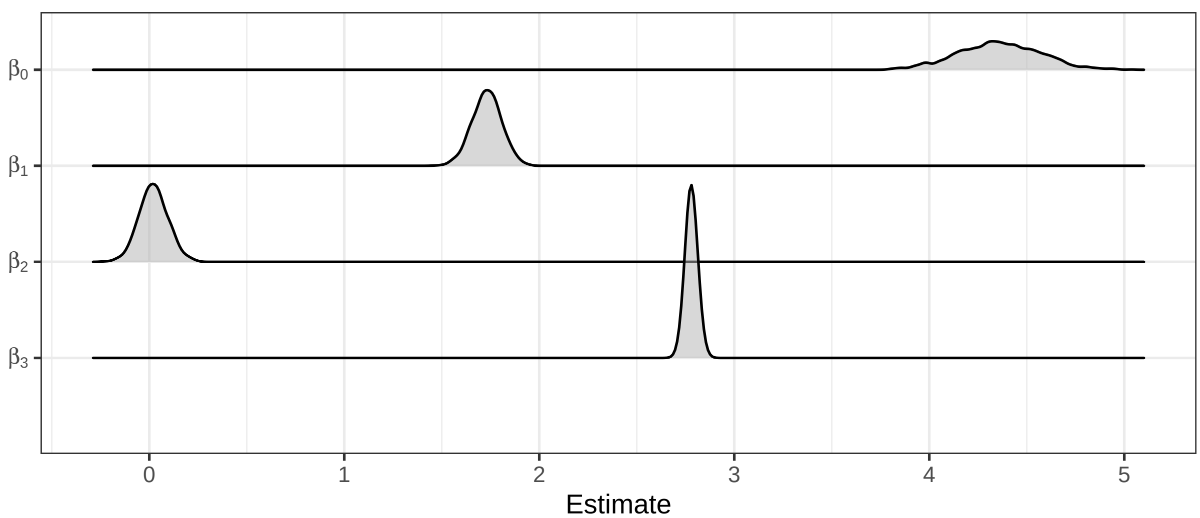

Figure 2: Distribution of the β coefficients

The spread of their distributions is very different. β_0 has the widest spread, followed by non-interaction terms β_1 and β_2. Then, the coefficient with the least spread is clearly the interaction term β_3

(c) Use the `simulate.reg()` function to simulate regression data using the default set of coefficients. Create a plot that shows the distribution of the $\hat{y}$s at $x_1 = 3.6$ and $x_2 = -1.9$. Make sure you set

`set.seed` to `F` so you can compare the output for differing samples of $\mathbf{x}$ and $\mathbf{Y}$. **What do you notice about the distribution?**

```
csim <- tibble(yhat = rep(NA, 1000))

for (i in 1:1000){


  dat <- simulate.reg(set.seed=F)

  mod <- data.frame(summary(lm(y ~ x1 + x2 + x1*x2, data = dat))$coefficients)

  betas <- mod$Estimate

  csim$yhat[i] <- betas[1] + 3.6*betas[2] - 1.9*betas[3] - 3.6*1.9*betas[4]

}

hist <- ggplot(data = csim, aes(x = yhat, y = after_stat(density))) +
  geom_histogram(bins = 15, color = "black", fill = "lightgrey") +
  theme_bw() +
  geom_hline(yintercept = 0) +
  xlab(bquote(hat(y))) +
  scale_x_continuous(breaks = c(-10, -9, round(median(csim$yhat),2), -8, -7), limits = c(-10,-7))

boxplot <- ggplot(data = csim, aes(x = "", y = yhat)) +
  geom_boxplot() +
  coord_flip() +
  xlab('') +
  ylab(bquote(hat(y))) +
  theme_bw() +
  scale_y_continuous(breaks = c(-10, -9, round(median(csim$yhat), 2), -8, -7), limits = c(-10,-7))

library(patchwork)
hist + boxplot +
  plot_layout(nrow = 2, heights = c(3, 1))
```

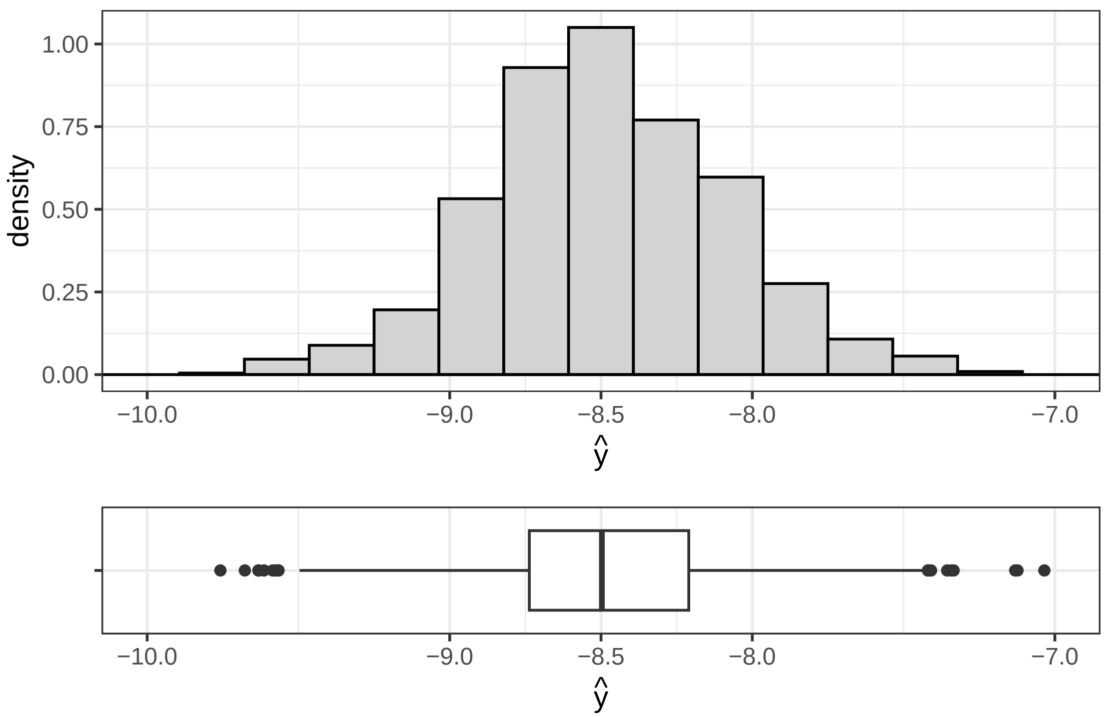

Figure 3: Distribution of $\hat{y}$'s at $x_1 = 3.6$ and $x_2 = -1.9$

The distribution of the $\hat{y}$ is Gaussian and centered at -8.5. The IQR is 0.53 [-8.74, -8.21] which is similar to the initial parameter σ.

(d) Describe why using t-based inference makes sense when conducting hypothesis tests or computing confidence intervals for a coefficient β_j, and the conditional expectation or prediction of Y?

2. Su et al. (2025) investigate how well-being of preschool teachers influences the effort they put into managing their emotions at work (emotional labor). They found that how committed a teacher is to their career (career commitment) plays a key role in linking well-being to emotional labor, and this connection gets stronger when they have more social support from colleagues or the community.

   As part of this work, they aimed to replicate findings that the level of social support moderates the impact of preschool teachers' career commitment on emotional labor. Below, we will assess this piece to the puzzle.

   Fit a regression model predicting `Emotional Labor` using `career commitment`, `social support`, and their interaction. Also include Gender, Age, and teaching experience.

   (a) Perform a residual analysis. Do you have any concerns about the model fit?
   (b) Perform an analysis of leverage/influence. Do you have any concerns about the model fit?
   (c) Provide the results of the overall F test. What does this tell you about the model you fit?
   (d) Based on the model fit, does the amount of social support impact the effect of career commitment on emotional labor?

3. Continue with the regression from Question 2.

   (a) Use `emmeans()` from the `emmeans` package for `R` (Lenth, 2024) to compute the estimated marginal means of `Emotional Labor` for low, average and high `career commitment` and `social support`. Report the results in a table and a figure.
   (b) Use `emtrends()` from the `emmeans` package for `R` (Lenth, 2024) to compute the slope for `career commitment` at low, average, and high `social support`. Report the results in a table and use `ggeffect()` from the `ggeffects` package for `R` (Lüdecke, 2018) to create a plot.
   (c) Use `pairs()` to compare the slopes for `career commitment` at low, average, and high `social support`. What does this tell us?
   (d) Use `johnson_neyman()` from the `interactions` package for `R` (Long, 2019) to evaluate which values of social suppport yield slopes for `career commitment` that are statistically discernible from zero. What does this tell us?

References

Lenth, R. V. (2024). *emmeans: Estimated Marginal Means, aka Least-Squares Means*. R package version 1.10.3.

Long, J. A. (2019). *interactions: Comprehensive, User-Friendly Toolkit for Probing Interactions*. R package version 1.1.0.

Lüdecke, D. (2018). ggeffects: Tidy data frames of marginal effects from regression models. *Journal of Open Source Software*, 3(26):772.

Su, X., Yu, D., Yu, Y., and Lian, R. (2025). Well-being and emotional labor for preschool teachers: The mediation of career commitment and the moderation of social support. *PLoS One*, 20(4):e0318027.